Key features

1 ∞

2

a As x increases, the function increases at a slower and slower rate.

b $(0, 0)$

c No

d No

e ∞

3

a

x	0	1	2	3	4	5	9
y	0	-1	-1.41	-1.73	-2	-2.24	-3

b No

c Decreasing function

d The function decreases at a slower and slower rate.

4

a

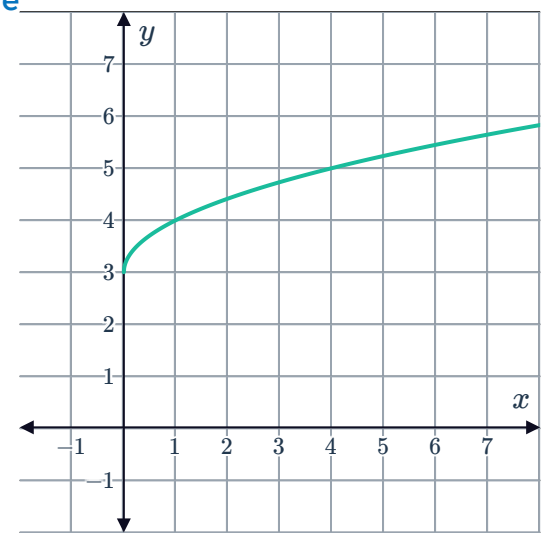
x	-5	-4	-3	-2	-1	0
y	2.24	2	1.73	1.41	1	0

b Decreasing

5

a No **b** ∞ **c** $y = 3$ **d** 0

e



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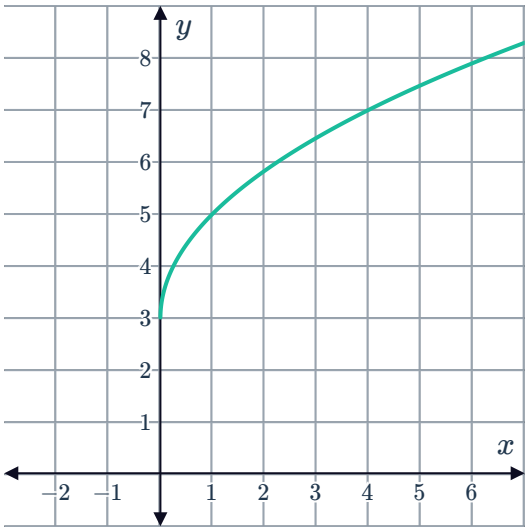
6

a Increasing

b More steep

c $(0, 3)$

d



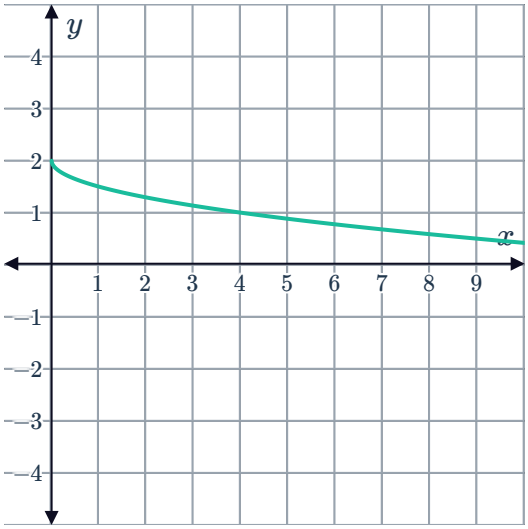
7

a Decreasing

b Less steep

c $(0, 2)$

d



Domain and range

8

a $x \leq 0$

b $y \geq 0$

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9

a

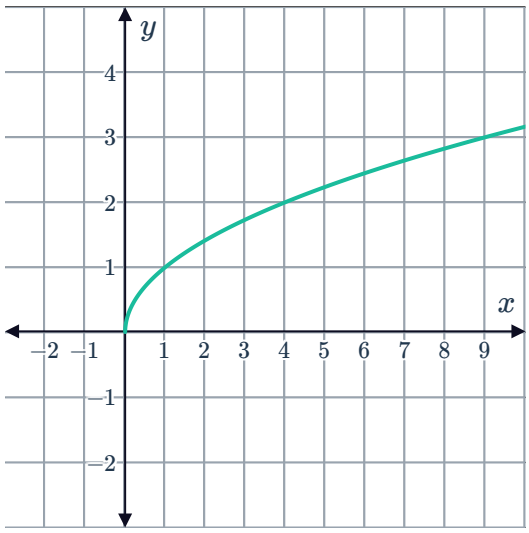
x	0	1	2	3	4	5	9
y	0	1	1.41	1.73	2	2.24	3

b $x \geq 0$

c $y \geq 0$

d ∞

e



10

a $x \geq 0$ **b** $y \leq 0$

11 $x \geq 0$ and $y \geq -2$

12

a $x \geq 5$ **b** $y \geq 0$ **c** Yes

13

a 6 **b** $x \leq 0$ **c** $y \geq 6$

14

a No **b** No **c** Yes **d** No

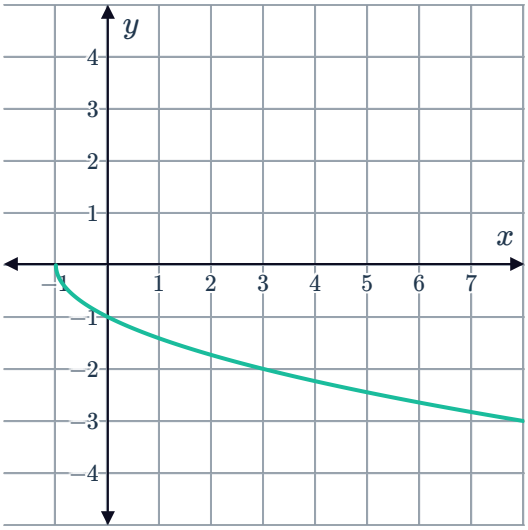
e Yes **f** No

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15

a

i

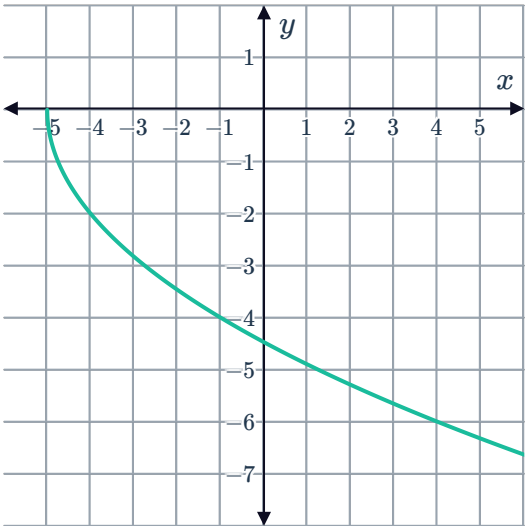


ii $x \geq -1$

iii $y \leq 0$

b

i



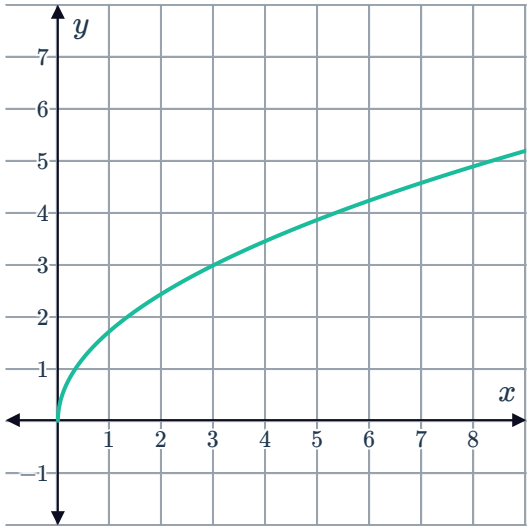
ii $x \geq -5$

iii $y \leq 0$

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c

i

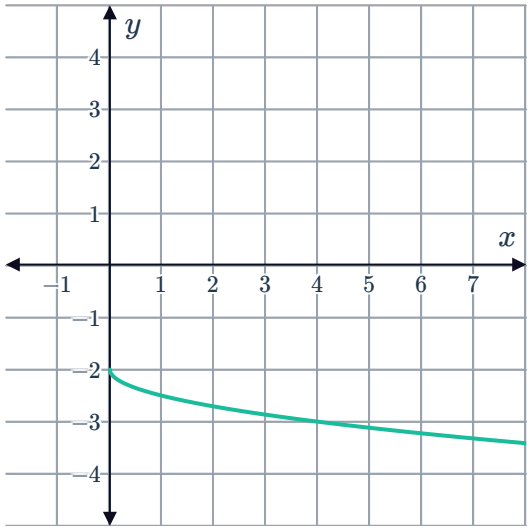


ii $x \geq 0$

iii $y \geq 0$

d

i



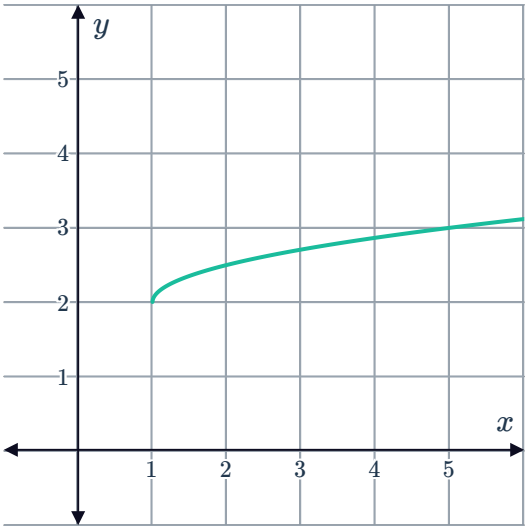
ii $x \geq 0$

iii $y \leq -2$

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e

i



ii $x \geq 1$

iii $y \geq 2$

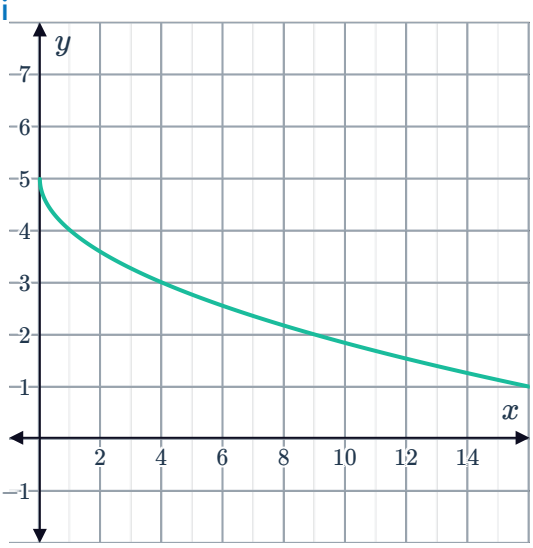
16

a

i $x \geq 0$

ii $y \leq 5$

iii

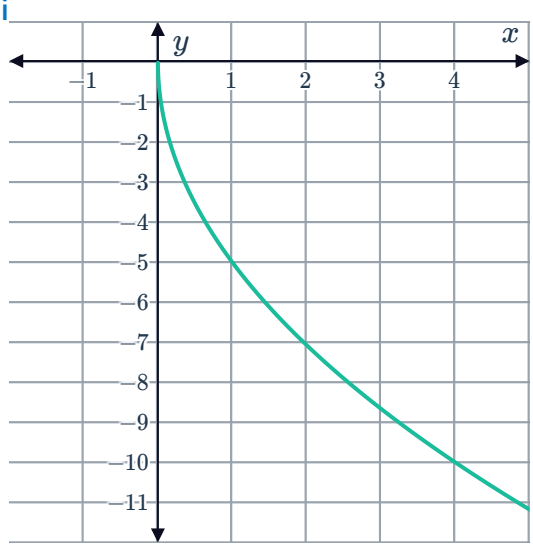


b

i $x \geq 0$

ii $y \leq 0$

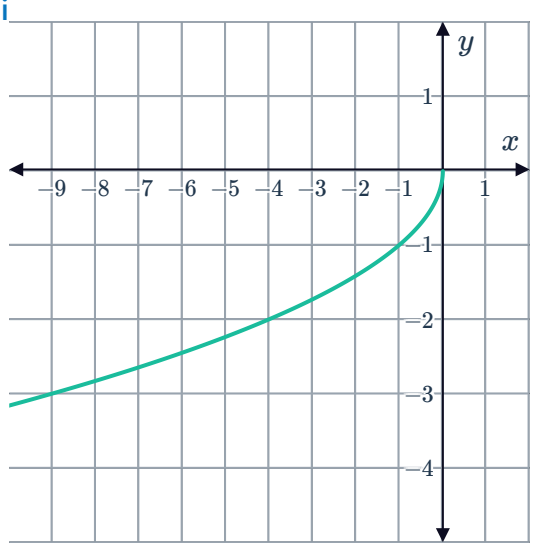
iii



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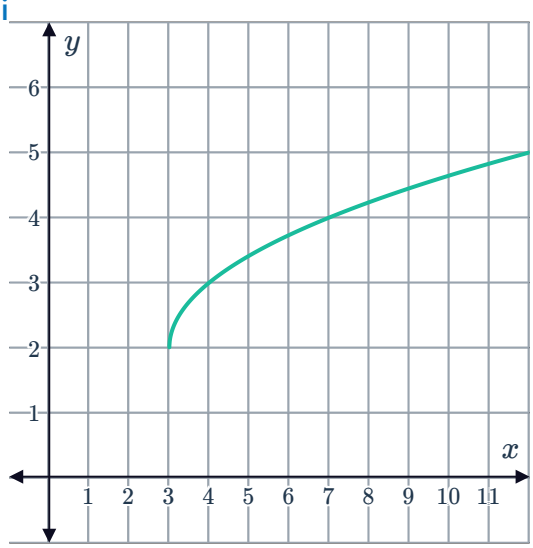
c

i $x \leq 0$ **ii** $y \leq 0$ **iii**



d

i $x \geq 3$ **ii** $y \geq 2$ **iii**



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17

a $x \geq 0$

b $x \geq 2$

c $x \leq 3$

d $x \leq 3$

e All real values of x.

18 $[2, \infty)$

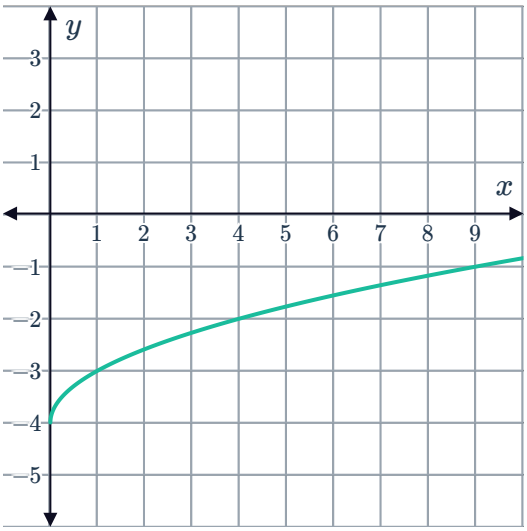
Transformation

19 $(0, 3)$

20

a The original function was translated down 4 units.

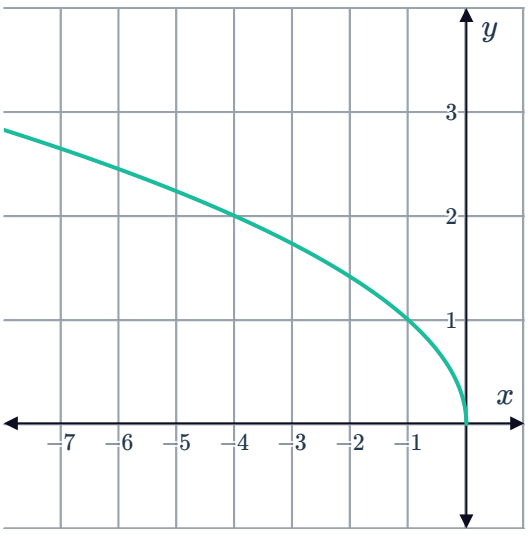
b



21

a

i

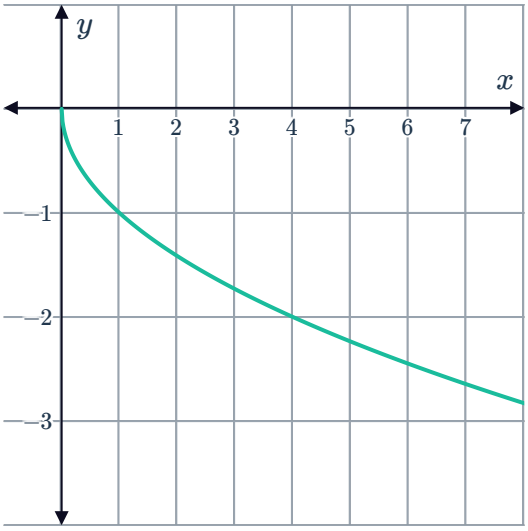


ii $y = \sqrt{-x}$

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b

i

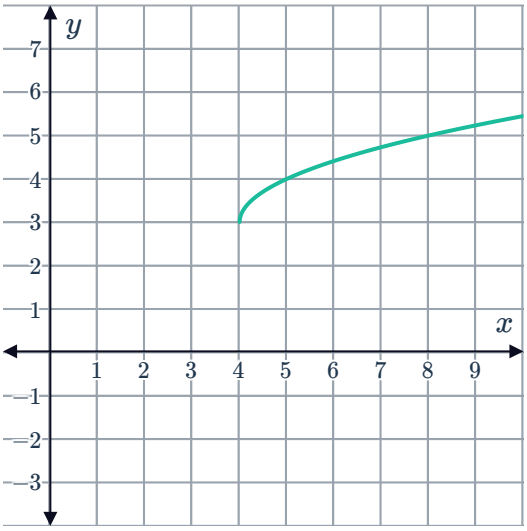


ii $y = -\sqrt{x}$

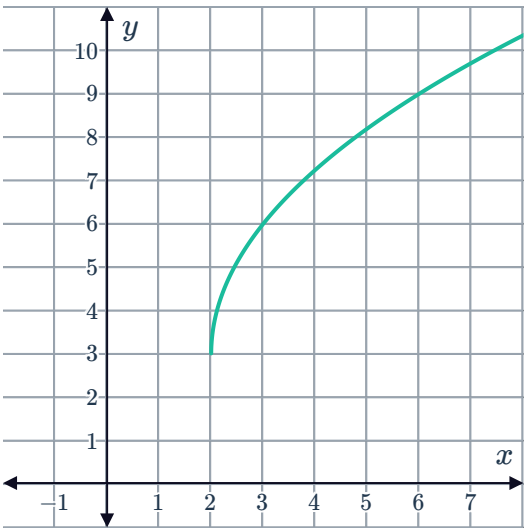
22

a Translate the graph 4 units to the right and 3 units up.

b



23



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